The Intrinsic Dimensionality of the Indian Stock Market: A Geometric Approach to Factor-Based Market Analysis

Problem Statement & Research Proposal

1 The Challenge: The "Curse of Dimensionality" in Financial Data

The Indian equity market, represented by broad indices like the NIFTY 500, is a high-dimensional system comprising hundreds of individual securities. For an analyst, this presents a significant challenge: tracking and modeling the co-movements of over 500 companies is computationally intensive and statistically unreliable. The sheer volume of data creates a phenomenon known as the "curse of dimensionality," making it difficult to distinguish meaningful market signals from noise.

The central problem this thesis addresses is the analytical complexity of this high-dimensional market space. The goal is to develop and validate a method to lower the dimension of the market analysis, moving from a space of hundreds of individual companies to a much smaller space of core, underlying factors. This reduced-dimension model will enable a more efficient and insightful way to analyze and explain the primary drivers of the Indian stock market.

2 Theoretical Motivation: The Geometry of High-Dimensional Spaces

Our intuition for geometry, honed in two or three dimensions, fails in high-dimensional spaces. The theoretical properties of such spaces provide a powerful motivation for our approach. As detailed in the foundational literature on high-dimensional data analysis (Blum, Hopcroft, and Kannan, 2020), high-dimensional vectors exhibit counter-intuitive properties such as distance concentration and inherent orthogonality [1].

2.1 Distance Concentration and Measure

The Law of Large Numbers states that the average of many independent random variables converges to its expected value. Since the squared Euclidean distance between two daily return vectors (y and z) is a sum of hundreds of squared differences, it can be viewed as a sum of d independent random variables:

$$|y - z|^2 = \sum_{i=1}^{d} (y_i - z_i)^2 \tag{1}$$

As the dimension d grows large, the probability of the sample average deviating from the true mean is bounded. This forces the total distance to "concentrate" around a single value.

- **Theoretical Context:** This phenomenon, known as the *Concentration of Measure*, renders distance metrics less meaningful in high-dimensional isotropic spaces, making it difficult to distinguish clusters or outliers [4].

2.2 Near-Orthogonality

In a high-dimensional random space, any two vectors drawn from a centered, isotropic distribution are, with overwhelming probability, nearly orthogonal (at a 90-degree angle) to each other.

- **Theoretical Context:** As shown in randomized algorithm literature, the volume of a high-dimensional unit sphere is concentrated near its "equator" relative to any fixed point, ensuring that random vectors are nearly perpendicular [5].
- **Financial Implication:** If market returns were purely independent random variables, the market's collective movement on one day would be uncorrelated with its movement on any other day. However, empirical observation of trends and volatility clustering contradicts this. The fact that we empirically observe structure (e.g., market-wide rallies) is strong evidence that stock returns are not independent random variables. This non-randomness implies that the market's data does not fill the entire 500-dimensional space but is constrained to a much lower-dimensional subspace.

3 Proposed Methodology: Subspace Projection

To identify this low-dimensional subspace, this thesis will employ a matrix factorization technique. The goal is to find the "best-fit k-dimensional subspace" for the cloud of data points representing daily market returns.

- **Theoretical Justification:** The feasibility of projecting high-dimensional data into a lower-dimensional space while preserving relative distances is guaranteed by the *Johnson-Lindenstrauss Lemma*, which provides the bounds for such dimensionality reduction [2].
- **Implementation:** Techniques such as Principal Component Analysis (PCA) or Random Projections can effectively recover this subspace if the intrinsic dimension k is small [3].
- **Research Steps:**
  1. **Data Collection:** Collect historical daily adjusted closing prices for NIFTY 500 constituents (e.g., 10 years) and transform them into a standardized log-return matrix.
  2. **Matrix Factorization:** Decompose the standardized return matrix into orthogonal vectors that define the principal directions of variation.
  3. **Intrinsic Dimension:** Analyze the variance captured by each component (e.g., using a scree plot) to determine the effective dimension k where the plot flattens.
  4. **Model Construction:** Retain only the top k principal components to construct the reduced factor model.

4 Expected Outcomes and Contribution

The primary outcome of this thesis will be a simplified, low-dimensional factor model of the Indian stock market.

- **Market Explanation:** Instead of analyzing 500 individual stocks, the market's behavior can be explained through the lens of a few (k) principal components. The first component is expected to represent the overall market movement (the "market factor"), while subsequent components will represent other systematic risk factors.

- **Analytical Efficiency:** This reduced model provides a computationally efficient framework for market analysis, risk assessment, and portfolio construction.

- **Novelty:** This thesis provides a novel framing by using the fundamental principles of high-dimensional geometry as the theoretical justification for the approach. It moves beyond a purely statistical application to a geometrically motivated one, bridging abstract geometric theory with practical financial problems.

References

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